

BAYANDAY, Russian Federation

WMO: 306270

Lat: 53.100N

Lon: 105.533E

Elev: 761

StdP: 92.51

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -36.0	(c) -33.6	(d) -38.6	(e) 0.1	(f) -35.8	(g) -36.0	(h) 0.1	(i) -33.6	(j) 6.0	(k) -20.2	(l) 5.3	(m) -22.2	(n) 1.9	(o) 30	(p) 0.629

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	27.9	17.5	26.0	16.7	24.2	16.0	19.1	24.9	18.1	23.1	17.1	22.4	2.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.2	13.5	21.2	16.2	12.6	20.5	15.2	11.8	19.6	57.9	25.1	54.4	23.1	51.4	22.6	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 5.7	(b) 4.8	(c) 4.2	(e) -37.7	(f) 31.1	(g) 4.5	(h) 2.2	(i) -41.0	(j) 32.7	(k) -43.7	(l) 34.0	(m) -46.2	(n) 35.2	(o) -49.5	(p) 36.8		
(a) 5.7	(b) 4.8	(c) 4.2	(e) -37.7	(f) 20.8	(g) 4.4	(h) 1.6	(i) -40.8	(j) 21.9	(k) -43.4	(l) 22.8	(m) -45.8	(n) 23.7	(o) -49.0	(p) 24.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.9	-21.5	-17.2	-8.0	1.6	8.4	15.3	17.8	15.1	8.0	-0.5	-12.0	-19.1
	DBStd	14.75	6.65	6.75	6.25	5.47	4.71	4.15	2.92	3.60	4.53	5.37	7.37	7.13
	HDD10.0	4628	976	762	558	254	89	6	0	4	91	326	659	903
	HDD18.3	7081	1234	995	817	501	311	108	44	109	309	583	909	1161
	CDD10.0	647	0	0	0	4	38	165	243	164	32	1	0	0
	CDD18.3	57	0	0	0	0	1	17	28	11	1	0	0	0
	CDH23.3	584	0	0	0	1	26	196	248	102	12	0	0	0
	CDH26.7	117	0	0	0	0	4	43	49	19	1	0	0	0
Wind	WSAvg	2.0	1.9	1.8	1.9	2.2	2.6	2.0	1.8	1.8	1.9	1.9	1.9	2.0
Precipitation	PrecAvg	375	6	5	6	13	25	58	95	86	42	18	10	9
	PrecMax	517	16	11	17	45	60	137	223	170	101	86	22	26
	PrecMin	222	1	0	0	1	1	13	28	18	11	4	2	1
	PrecStd	63	3	3	4	9	14	29	48	39	22	16	4	6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.3	-0.6	8.9	20.5	26.4	30.5	30.3	28.9	25.1	16.9	5.7	-2.8
		MCWB	-6.3	-3.4	2.9	9.2	13.4	18.3	19.1	17.7	15.1	8.5	2.1	-4.7
	2%	DB	-8.4	-3.1	5.4	16.4	22.7	27.6	28.0	25.9	20.9	12.5	2.1	-6.0
		MCWB	-9.7	-5.4	0.7	7.4	11.9	17.0	18.2	16.7	13.3	6.2	-0.6	-7.0
	5%	DB	-10.8	-5.4	2.8	13.1	20.2	25.6	26.0	23.8	18.0	9.5	-0.1	-8.3
		MCWB	-11.6	-7.2	-0.9	5.6	10.3	16.1	17.5	16.2	11.7	4.5	-2.0	-9.0
	10%	DB	-12.8	-7.9	0.7	10.2	17.4	23.5	24.2	21.8	15.2	7.0	-1.9	-9.9
		MCWB	-13.4	-9.3	-2.3	4.0	8.7	15.0	17.1	15.4	10.1	2.9	-3.4	-10.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.1	-3.3	3.1	9.7	14.5	20.3	21.1	19.0	15.7	9.0	2.1	-4.6
		MCDB	-5.3	-0.9	8.3	19.1	24.6	27.5	27.2	25.0	23.0	16.3	5.6	-3.2
	2%	WB	-9.5	-5.3	0.9	7.6	12.5	18.1	19.7	17.7	14.1	6.6	-0.4	-7.0
		MCDB	-8.4	-3.3	5.0	15.4	21.8	24.7	25.5	23.8	20.0	11.9	1.6	-6.1
	5%	WB	-11.5	-7.3	-0.7	5.9	10.7	16.9	18.7	16.8	12.3	4.8	-1.8	-8.9
		MCDB	-10.8	-5.5	2.5	12.9	18.9	23.8	23.8	22.5	16.7	9.3	-0.1	-8.3
	10%	WB	-13.4	-9.3	-2.1	4.2	9.2	15.7	17.8	15.9	10.8	3.1	-3.3	-10.4
		MCDB	-12.8	-7.9	0.7	9.7	16.3	22.1	22.1	20.8	14.8	6.5	-1.9	-9.9
Mean Daily Temperature Range	5% DB	MDBR	6.5	8.2	9.7	10.4	11.6	11.5	10.3	10.1	10.0	9.3	7.5	6.8
		MCDBR	7.5	8.7	10.8	14.7	15.6	14.7	13.9	13.4	13.9	13.1	8.6	8.0
		MCWBR	6.7	7.1	7.4	8.5	8.5	7.0	6.3	6.2	7.3	8.0	6.7	7.2
	5% WB	MCDBR	7.5	8.5	10.2	14.3	14.5	12.7	11.2	11.5	12.6	12.6	8.5	7.9
		MCWBR	6.7	6.9	7.2	8.5	8.3	6.8	5.9	5.8	7.0	7.8	6.7	7.2
Clear-Sky Solar Irradiance	taub	0.259	0.284	0.312	0.356	0.405	0.431	0.418	0.371	0.325	0.302	0.280	0.259	
	taud	2.105	2.097	2.125	2.254	2.158	2.172	2.256	2.373	2.460	2.428	2.231	2.085	
	Ebn at Noon	706	807	865	873	843	822	827	851	856	799	681	619	
	Edh at Noon	70	98	119	120	140	140	126	106	86	71	62	59	
All-Sky Solar Radiation	RadAvg	1.21	2.34	3.91	4.95	5.51	5.87	5.39	4.66	3.38	2.12	1.19	0.84	
	RadStd	0.06	0.08	0.15	0.32	0.35	0.33	0.44	0.20	0.30	0.14	0.07	0.06	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)		N/A	N/A	-1.97	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page